

INTRODUCTION TO AI AND APPLICATIONS

Module 1

Module 1:

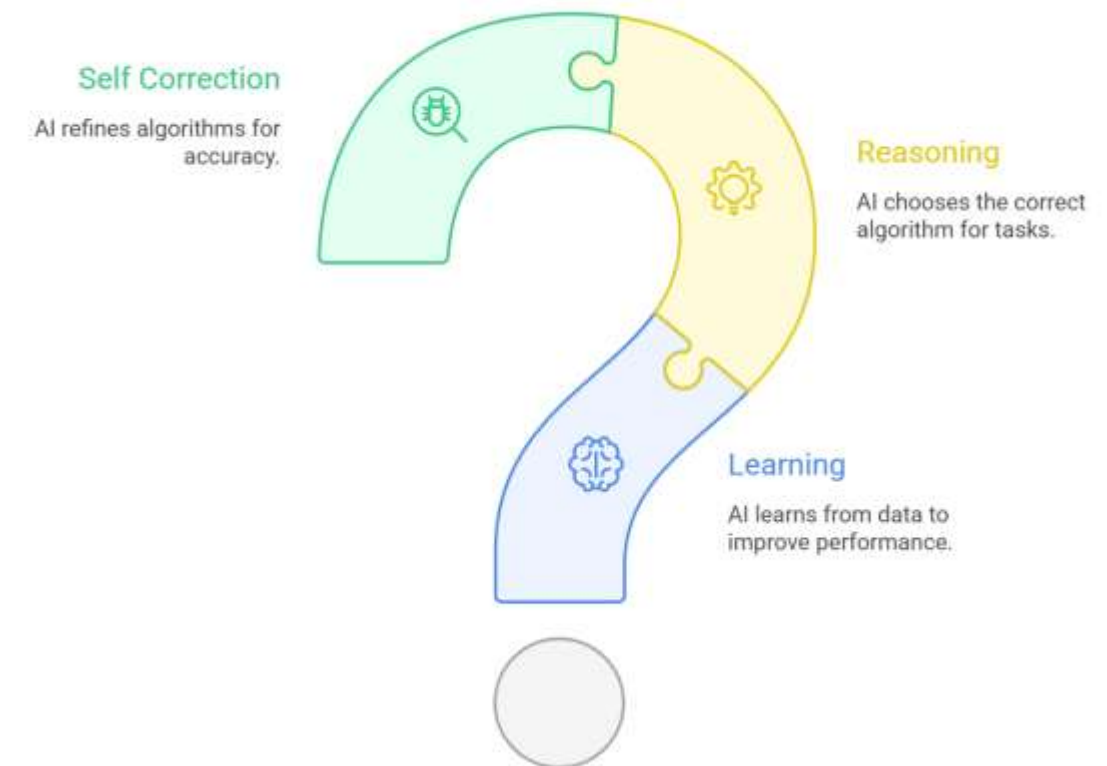
Definition of Artificial Intelligence

- **Artificial Intelligence (AI)** is the science and engineering of making intelligent machines, especially intelligent computer programs, *John McCarthy (2004)*.
- **Historical Background:** The concept of AI was first explored in 1950 by Alan Turing, a British mathematician and computer scientist, who asked the question "Can machines think?" This was a groundbreaking idea and led him to propose the Turing Test.
- **The Role of John McCarthy (2004):** Later, in 2004, John McCarthy defined AI as the science and engineering of making intelligent machines—basically, how we can program computers to act smart like humans.
- **A Researcher's Perspective:** For researchers, AI refers to a set of algorithms (step-by-step instructions) that help a machine make decisions and act without being explicitly told what to do each time.
- **Understanding AI from a Simple View:** Think of AI as machines or software that are designed to learn from their environment, just like humans learn from their experiences. For example, an AI program can be trained to recognize your face.
- **Everyday AI Examples:** Some well-known examples of AI include chess-playing computers or self-driving cars. These systems depend on deep learning (a type of AI that mimics the human brain) and natural language processing (helping computers understand human language, like Siri or Google Assistant).

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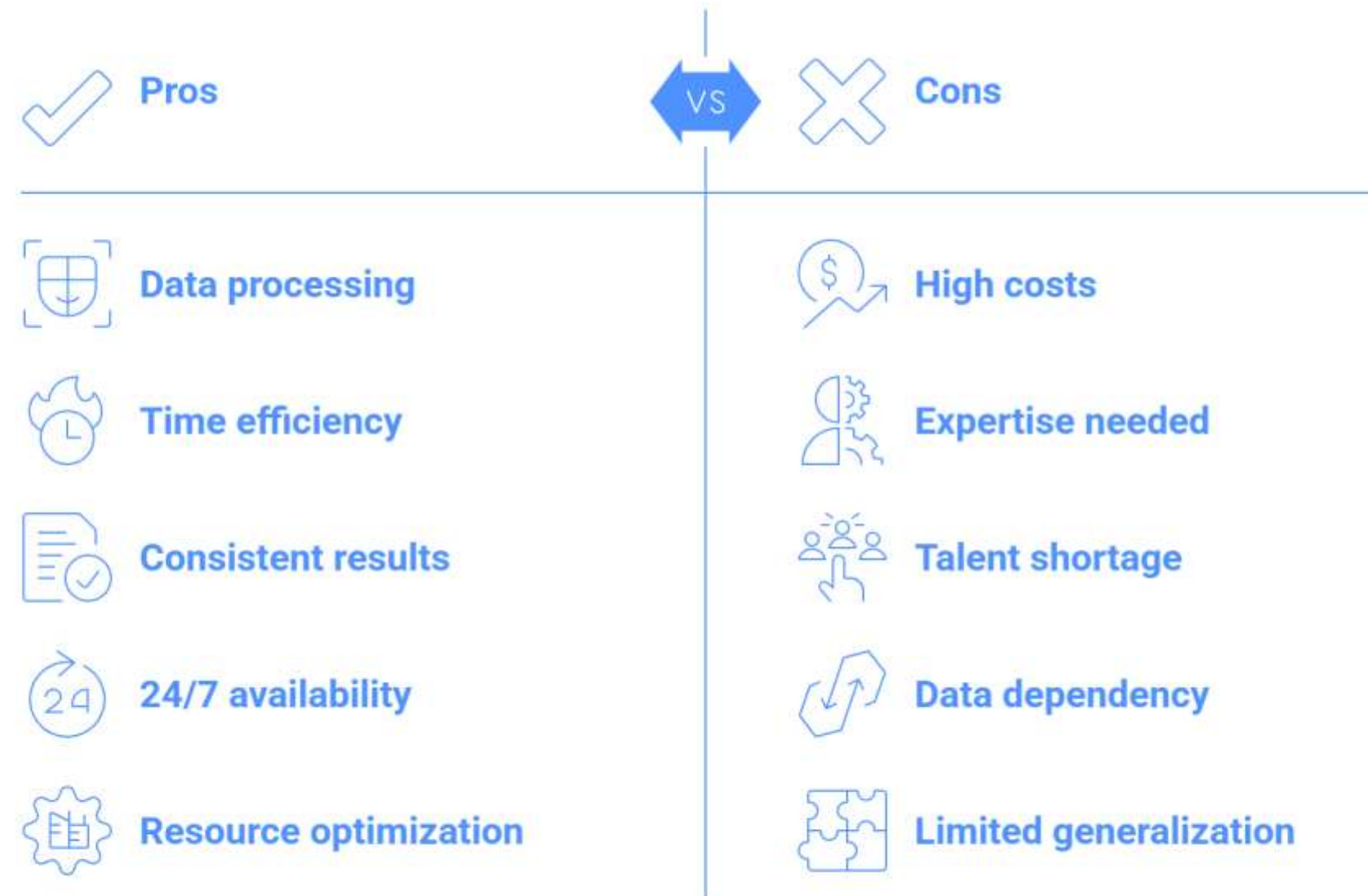
How Does AI Work?

- AI works by processing large datasets, recognizing patterns, and making decisions using algorithms.
- It involves learning, reasoning, and self-correction:
 - a. Learning - AI learns from data
 - b. Reasoning - AI chooses the correct algorithm
 - c. Self-Correction - AI refines algorithms for accuracy



Module 1:

Advantages and Disadvantages of AI

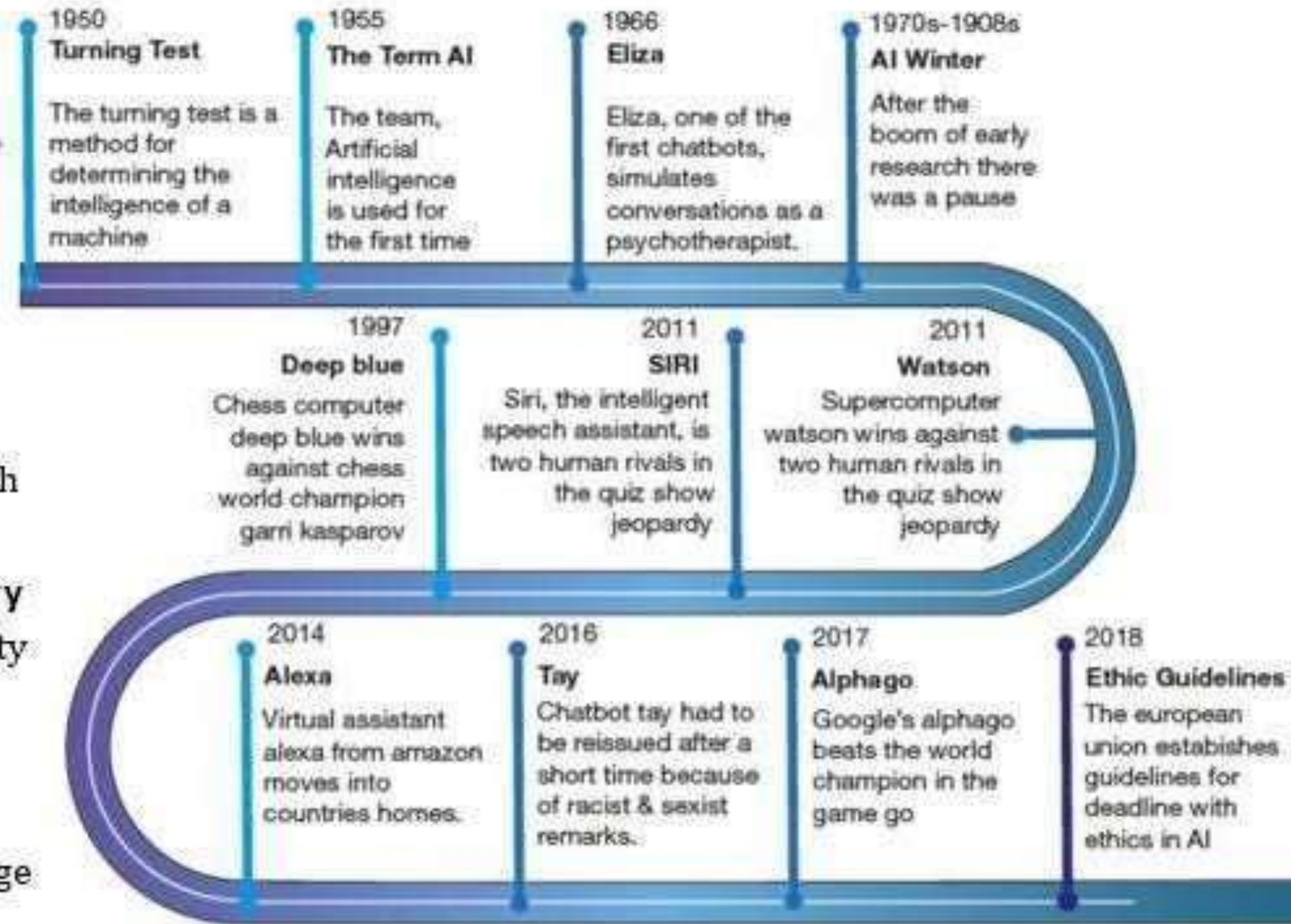


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Module 1: Introduction to Artificial Intelligence

History of AI

- **1943 - First Neural Network Model Proposed:** By Warren McCullough and Walter Pitts to lay the foundation for artificial neural networks and machine learning.
- **1950 - Turing Test Introduced:** By Alan Turing to measure a machine's ability to exhibit intelligent behavior indistinguishable from humans.
- **1956 - John McCarthy Coins the Term 'Artificial Intelligence':** By John McCarthy during the Dartmouth Conference, marking the birth of AI as a formal field.
- **1997 - IBM Deep Blue Defeats Chess Champion Garry Kasparov:** By IBM's team to demonstrate AI's capability in strategic decision-making through computational power.
- **2011 - Siri Introduced by Apple:** By Apple to revolutionize personal assistants using natural language processing and AI for everyday tasks.

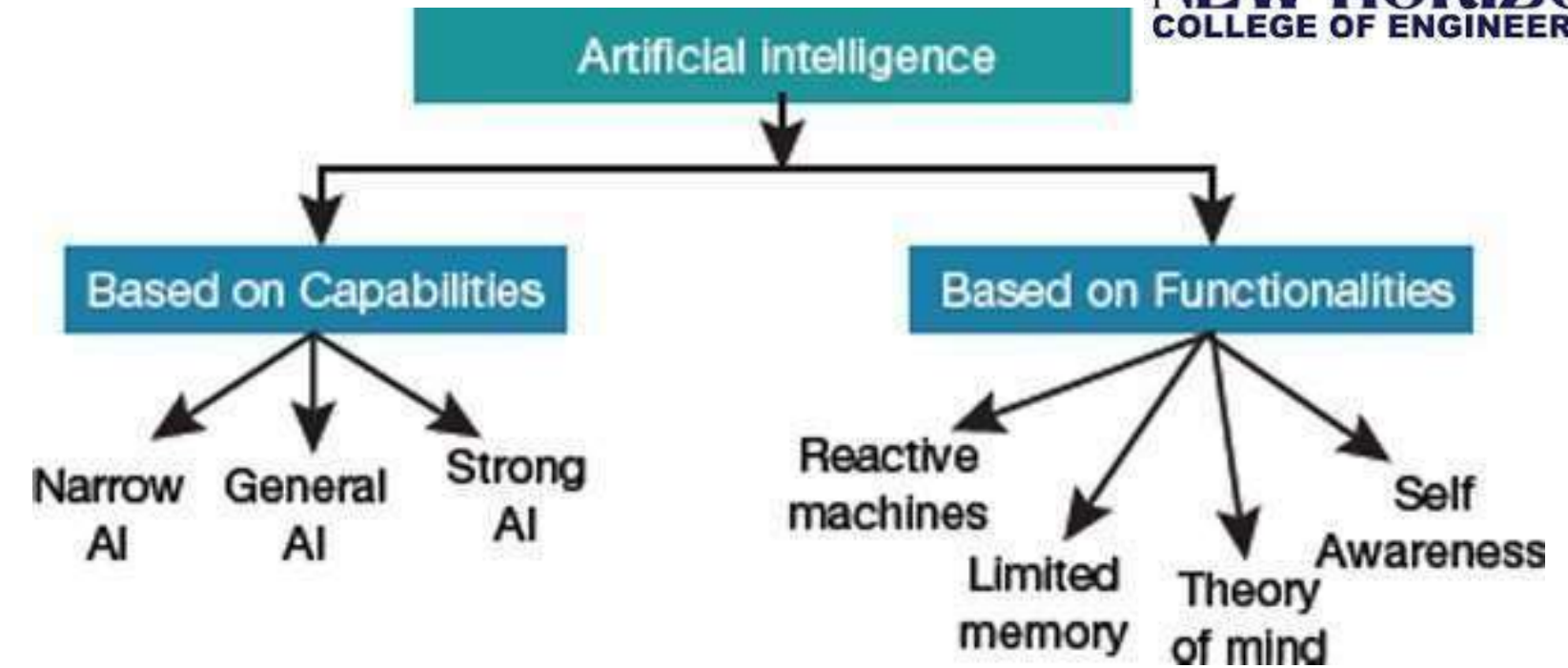


Module 1:

Types of AI

AI can be categorized into:

- **Weak AI (Narrow AI)**
 - Weak AI, also known as narrow AI, is designed to do one specific task.
 - Siri and Alexa are examples.
- **Strong AI (Artificial General Intelligence)**
 - Strong AI, also called Artificial General Intelligence (AGI) or Superintelligence (ASI), tries to mimic human thinking.
 - Can perform tasks it hasn't been specifically trained for.
 - Requires abilities like visual perception, speech recognition, decision-making, and language translation.



Module 1:

Types of AI

AI can be categorized into:

Progression of AI Capabilities

Based on Functionalities



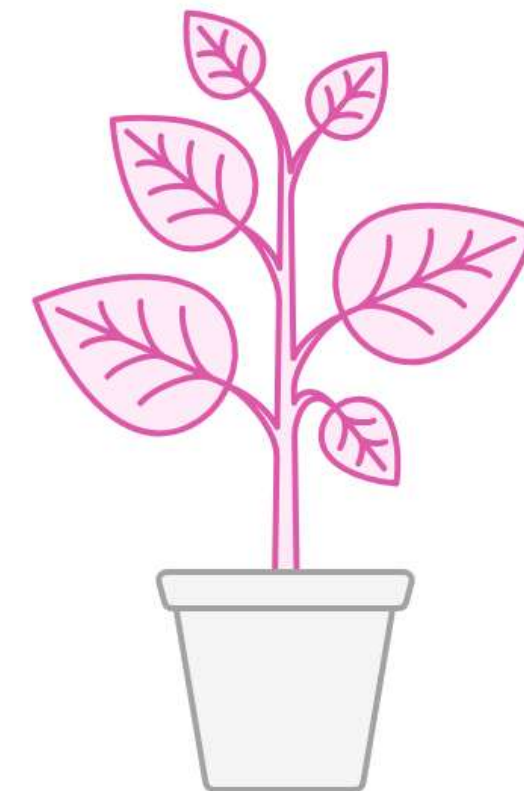
Reactive
Machines



Limited Memory



Theory of Mind



Self-Awareness

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Module 1:

Types of AI

1. Weak AI (Narrow AI)

Weak AI, also known as narrow AI, is designed to do one specific task.

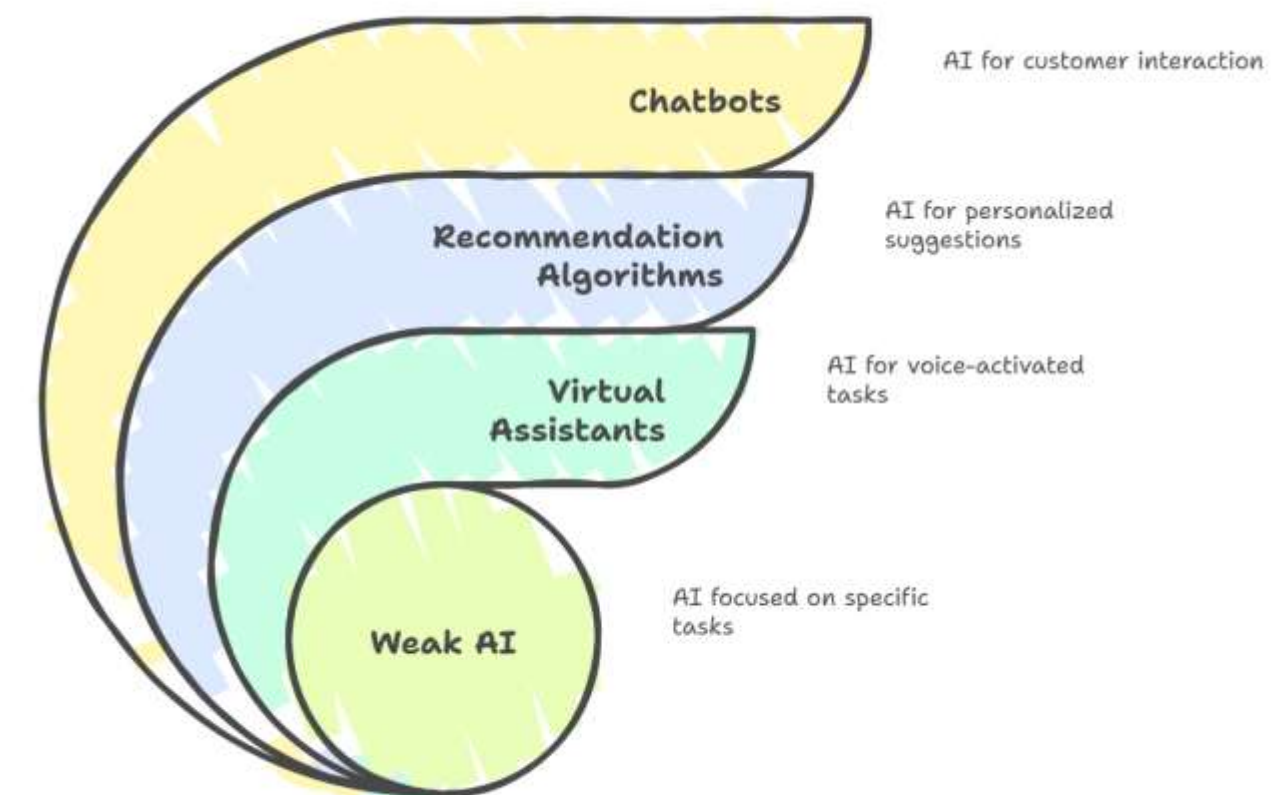
Examples:

Siri and Alexa are examples. When you tell Alexa to play a song, it does so because it's trained to understand that specific command.

Other examples include weather forecasting, predicting stock prices, and Google search.

How it works: These systems are great at doing one thing really well, but they don't work outside the specific task. For example, Alexa can't drive a car; it's just built for voice commands.

Why it's important: Weak AI has helped make many tasks easier and more efficient, and it is the most common type of AI in use today.



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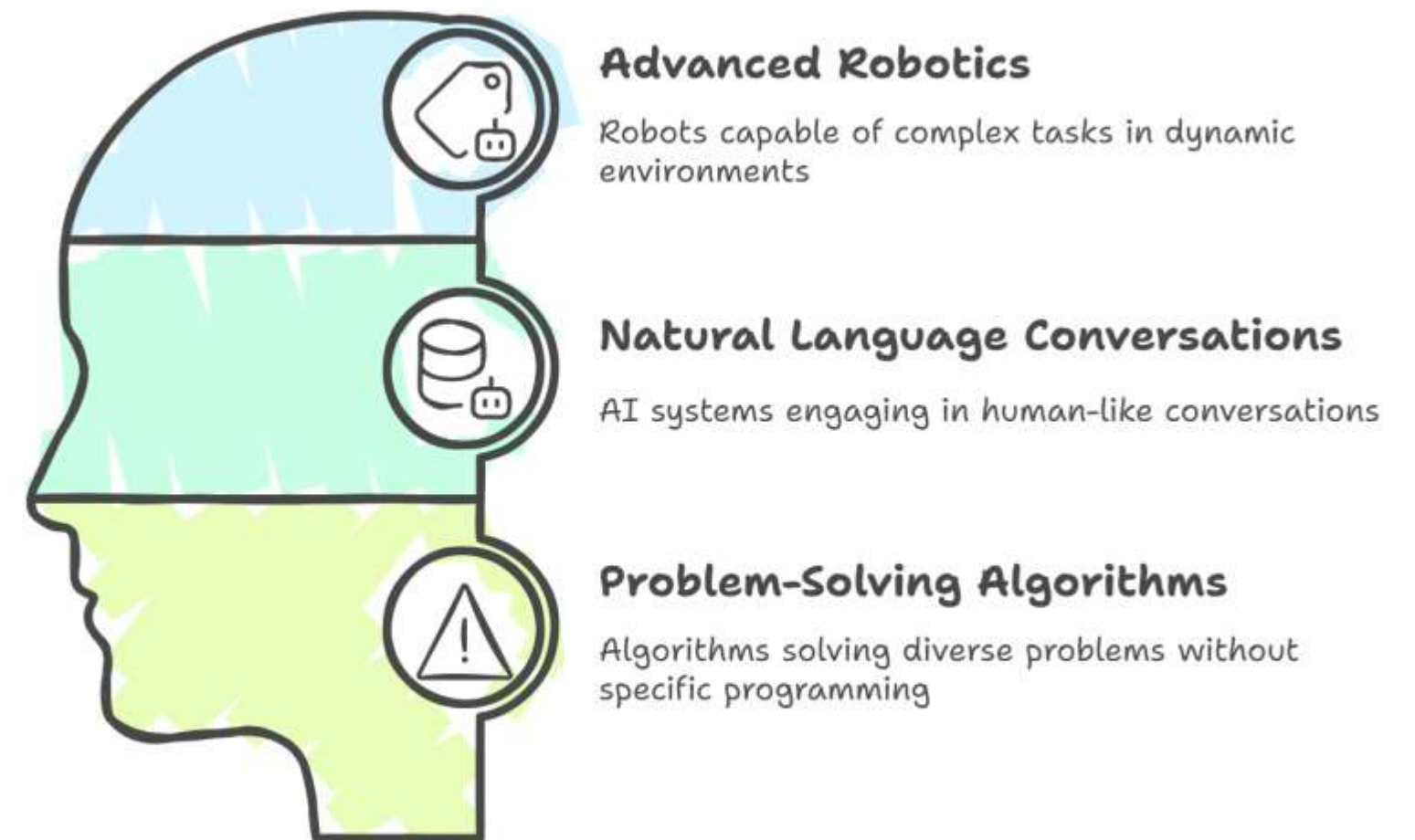
Types of AI

2. Strong AI (Artificial General Intelligence – AGI)

- Strong AI, also called Artificial General Intelligence (AGI) or Superintelligence (ASI), tries to mimic human thinking.
- Can perform tasks it hasn't been specifically trained for.
- Requires abilities like visual perception, speech recognition, decision-making, and language translation.

Future potential: Experts believe that Strong AI might one day surpass human intelligence, but it's not expected to happen anytime soon.

Concerns: While some fear that Strong AI could be dangerous, experts say we don't need to worry about it in the near future, as it's still far from becoming a reality.

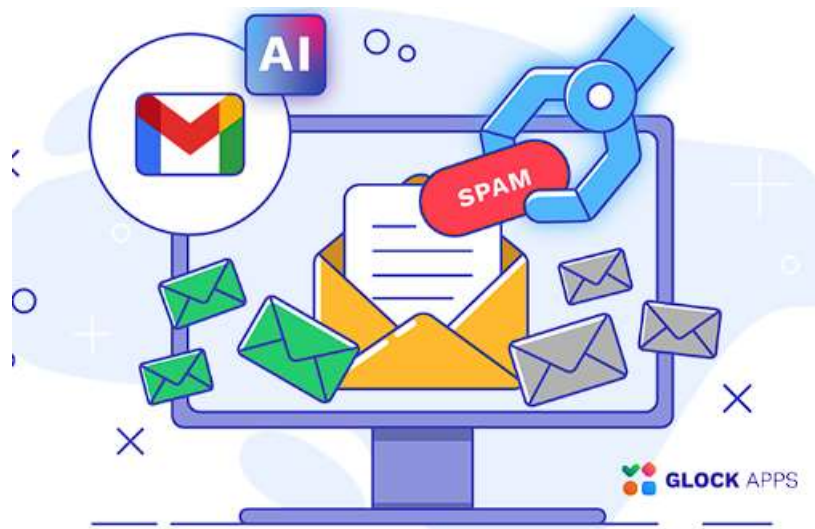


Module 1:

Types of AI

3. Reactive Machines

- **Reactive Machines** are the simplest type of AI that react to situations based on immediate input, but they have no memory or ability to learn from past experiences.
 - **Examples:** IBM's Deep Blue (chess-playing computer) is a reactive machine. It makes decisions based on the current state of the game but doesn't remember past games.



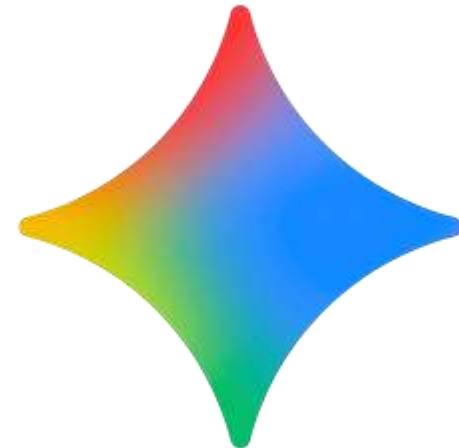
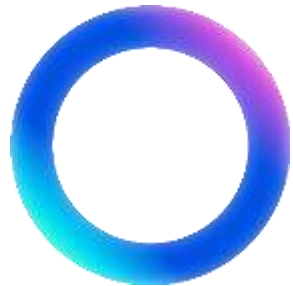
[Gmail's Biggest Security Update: AI-Powered Spam Filter | GlockApps](#)



[Deep Blue: the computer that revolutionized the world of chess](#)



[This Grand Theft Auto 5 Mod Lets AI NPCs Insult You - Decrypt](#)



Module 1:

Types of AI

4. Limited Memory

- What it is: Limited memory AI systems can remember data for a short time and use it to make decisions, but they don't keep data permanently.
 - **Examples:**
 - Autonomous vehicles use limited memory to track information like speed of nearby cars, distance between cars, and speed limits to navigate safely.
 - AlphaGo, the AI that defeated the world champion in the game Go, also used limited memory to play and improve during the game.
 - These systems learn and improve continuously by analyzing new data and adjusting based on feedback.
 - **Key Models:**
 - **Reinforcement Learning:** AI learns by trial and error, improving over time.
 - **Long Short-Term Memory (LSTM):** AI uses past data to predict the next step, but it focuses more on recent data.
 - **Evolutionary GANs (E-GAN):** The AI evolves over time, using data and feedback to make better decisions and predict outcomes.

Module 1:

Types of AI

5 Theory of Mind

- The Theory of Mind in AI aims to create machines that can understand thoughts, emotions, and memories— just like humans.
- **How it works:** AI would need to understand feelings and emotions that influence decisions. These machines would make choices by considering both reason and emotional context.

Current Status:

Requires major advances in emotional and social intelligence to model and predict human mental states. Some researchers are exploring hybrid neuro-symbolic AI to blend deep learning with structured reasoning.



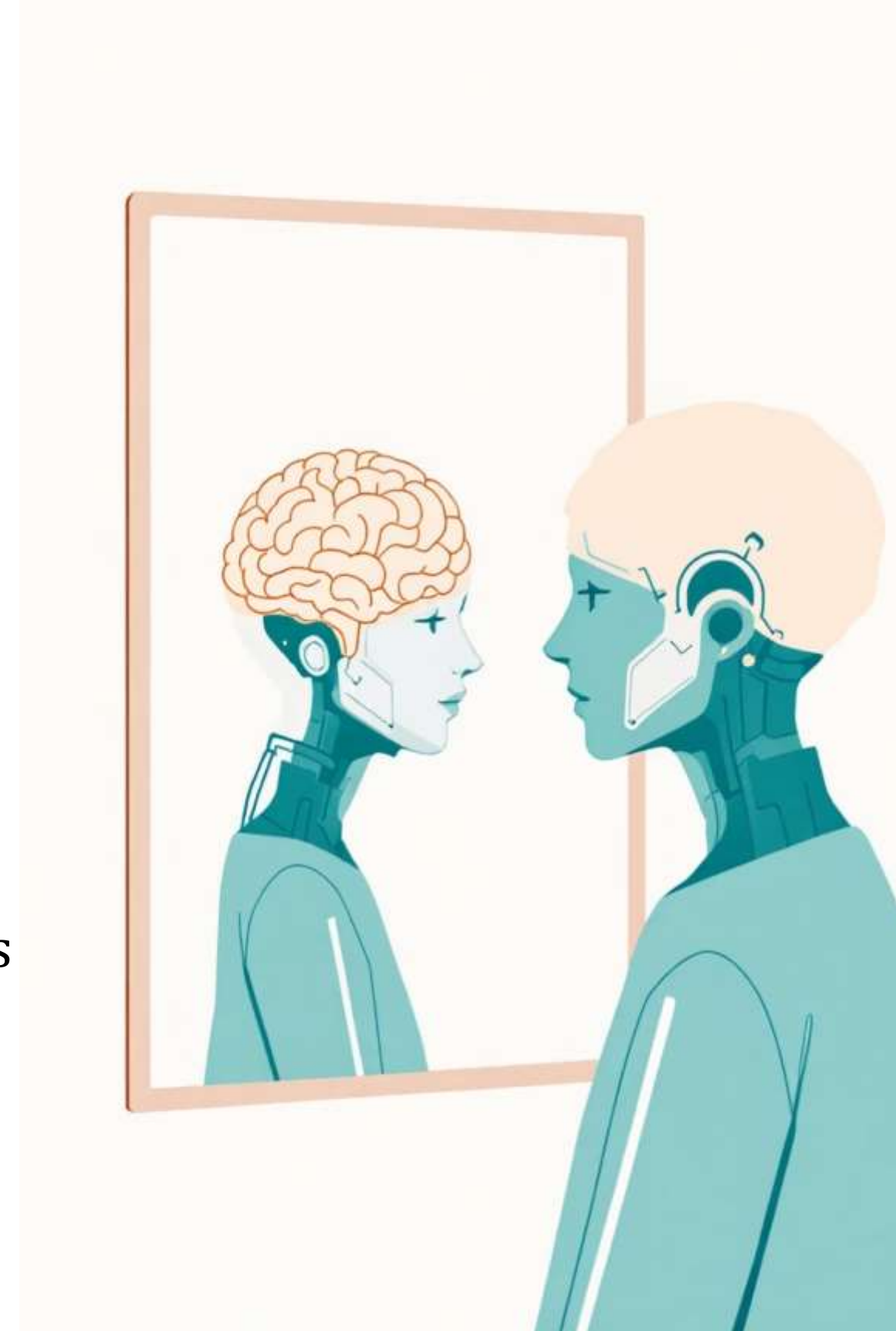
Module 1:

Types of AI

6 Self-Awareness

- **What it is:** Self-awareness in AI means machines that have a human-level consciousness—they can understand their own existence and feelings.
- **How it works:** These machines would not only understand what someone says but also how they feel based on the way they communicate. They could learn and adapt their responses to the emotions of others.

Current Status: It is currently a purely theoretical concept, as researchers do not yet fully understand how consciousness works in humans, making it impossible to replicate in a machine.



machine rights and slavery

Module 1:

Is AI the Same as Augmented Intelligence and Cognitive Computing?



Augmented Intelligence

A type of weak AI that assists humans to improve tasks and decision-making processes.

Example: Automatically highlighting important information in reports or documents.

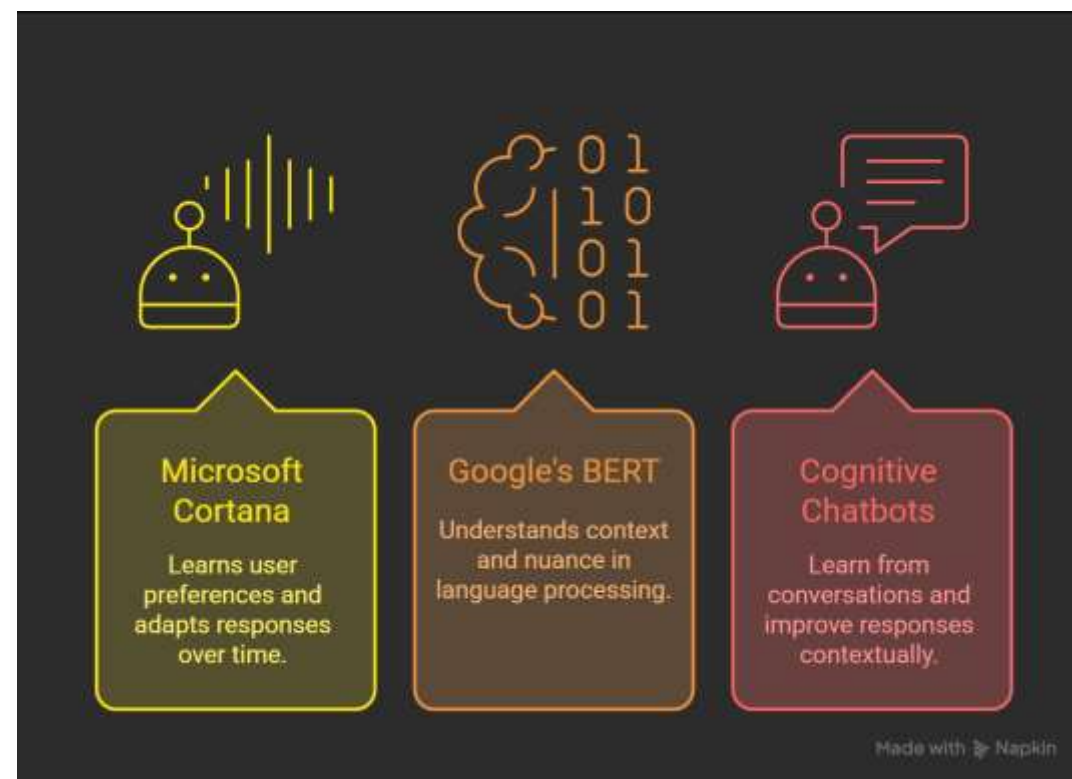
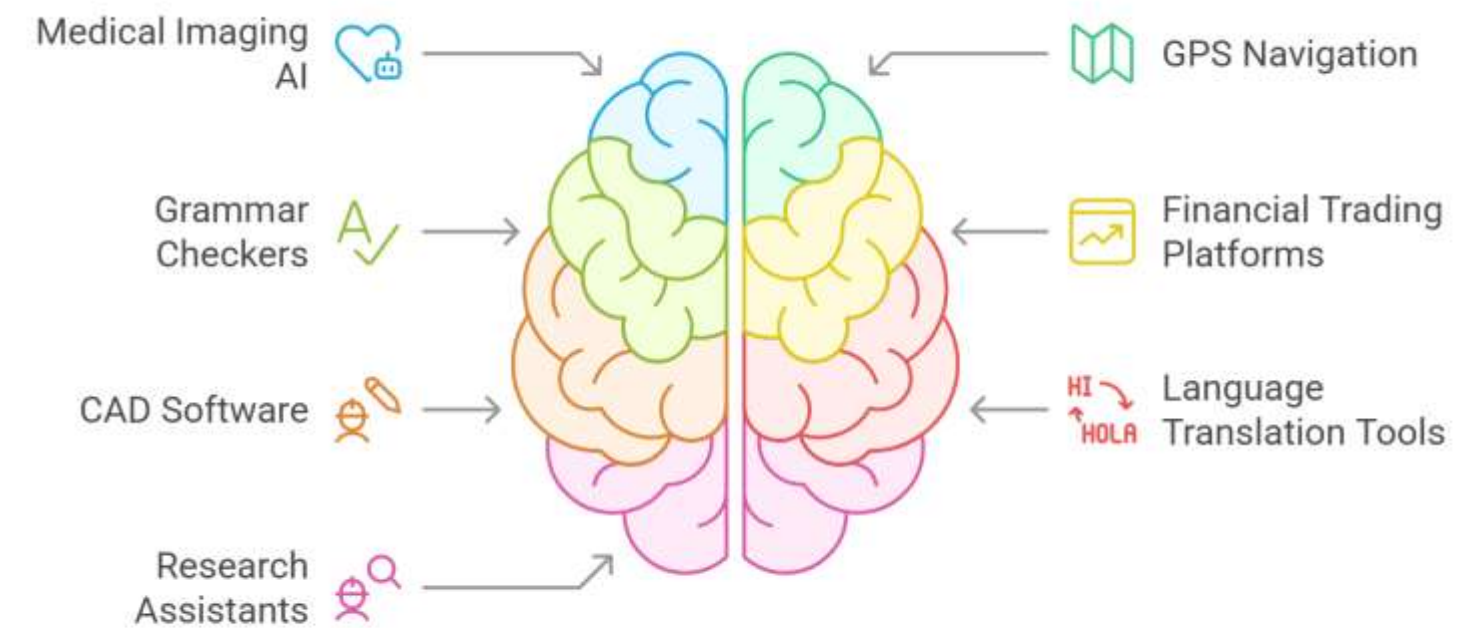
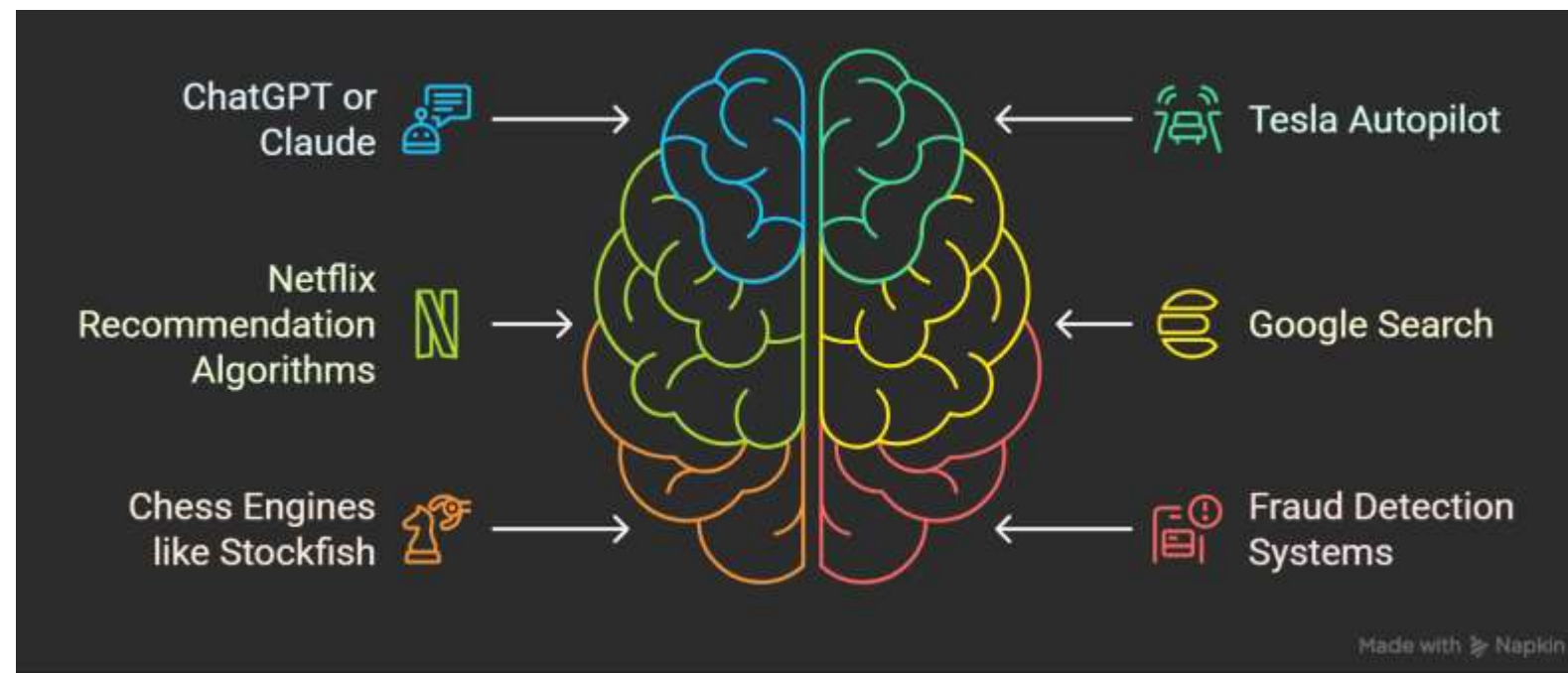
Cognitive Computing

Systems or products that mimic human thought processes to enhance decision-making capabilities and problem-solving skills.

AI in Machines

AI enables machines to simulate human intelligence through learning, sensing, processing, and reacting to environmental information.

Module 1:



Module 1:

Machine Learning (ML)

- Machine Learning is a branch of AI that teaches machines to learn from data and make decisions without being explicitly programmed.

01

Finding Patterns

ML algorithms analyze vast amounts of data to identify meaningful patterns and relationships that might not be obvious to humans.

02

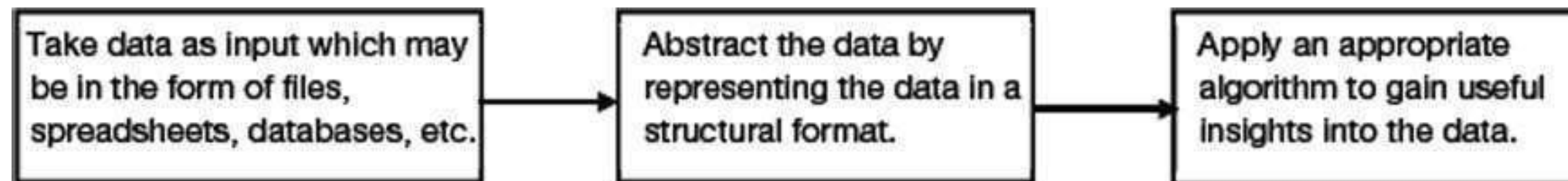
Learning from Experience

Machines improve automatically by learning from their past outputs and outcomes, getting smarter with each interaction.

03

Self-Correction

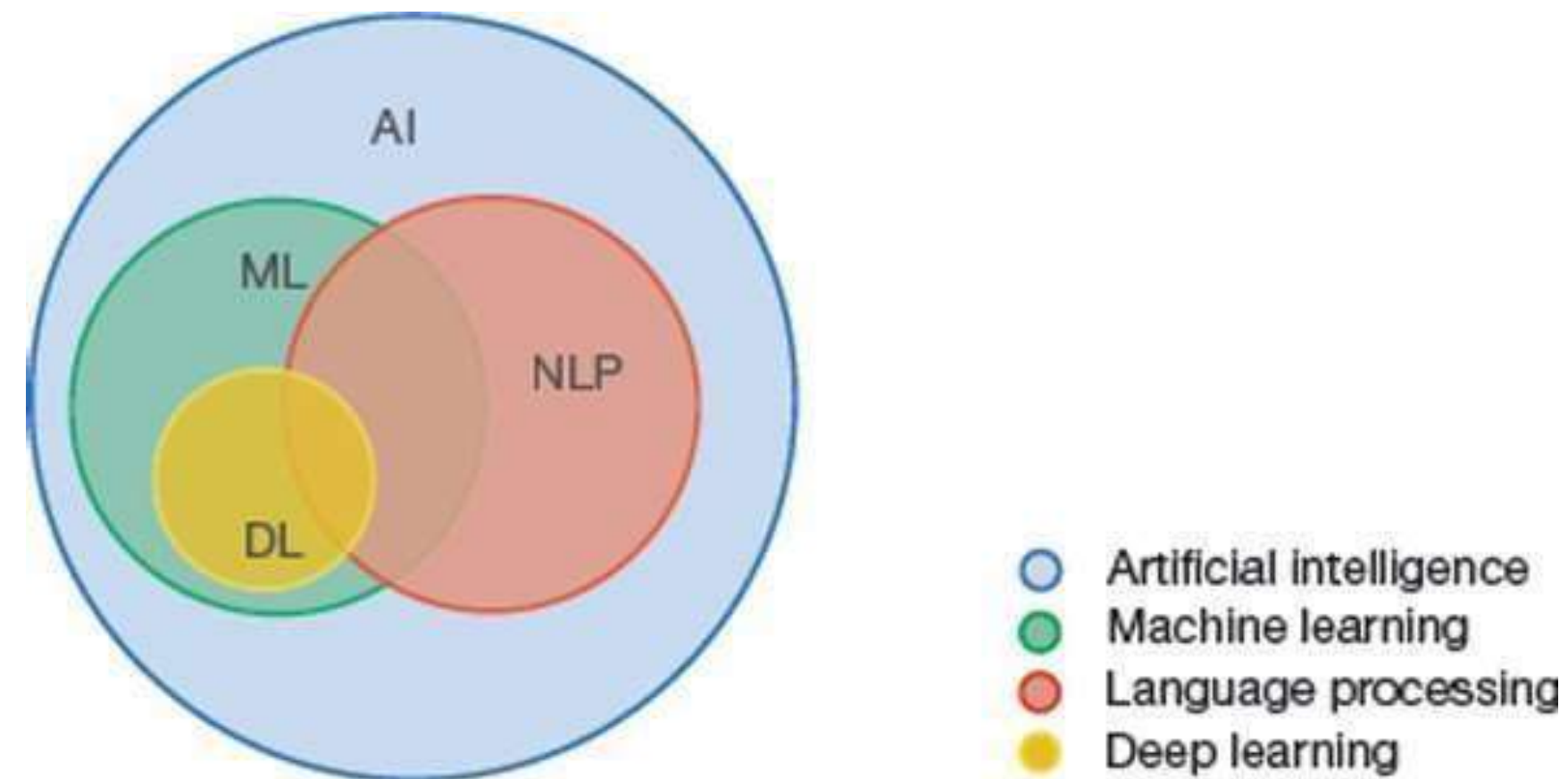
When the machine makes a mistake, it adjusts its approach and learns to improve accuracy over time through continuous feedback.



- Real-life Example:** Number series problem - finding the missing number in 10, 20, 30, 40 → 50, just like humans naturally do through pattern recognition.

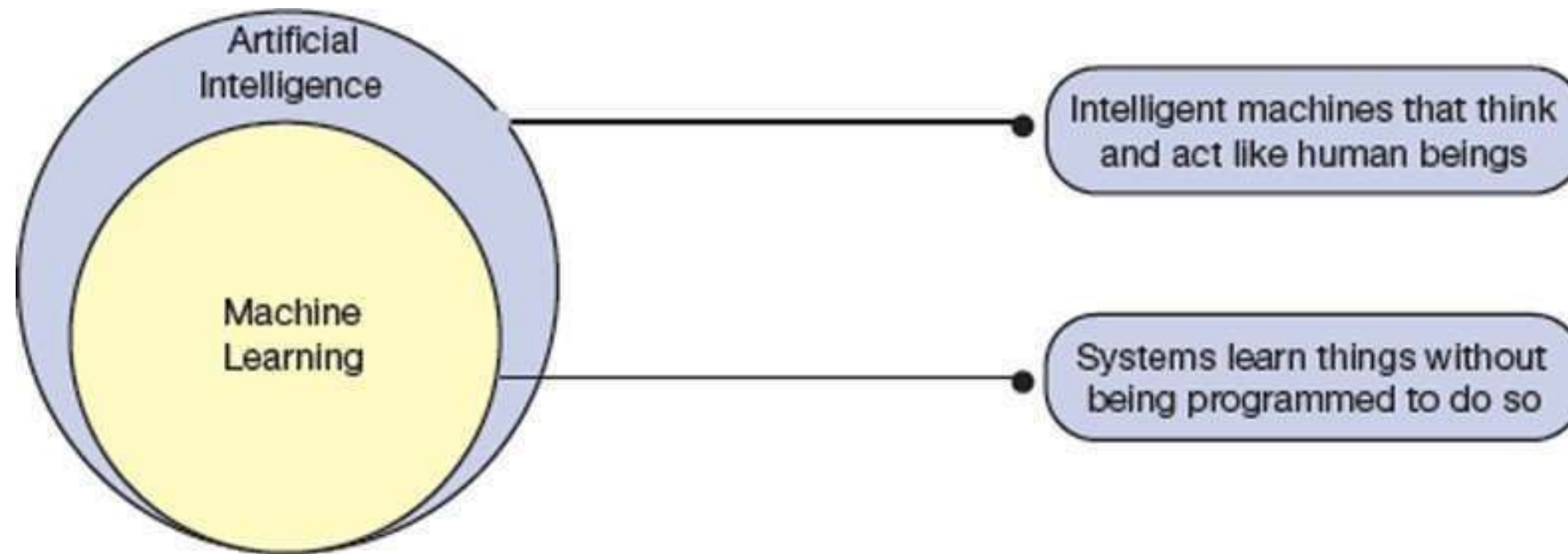
Module 1:

Relationship between Artificial Intelligence, Machine Learning, Deep Learning and Natural Language Processing



Module 1:

HOW IS AI RELATED TO MACHINE LEARNING?



AI vs ML:

- AI is the larger goal (the superset) that aims to make machines intelligent.
- Machine Learning is a subset of AI, used specifically for learning from data to make decisions.
- In simple terms, ML helps achieve AI by making machines learn and adapt.

Example:

- If you want a robot that can see, talk, walk, and learn, you would use AI because it requires many different technologies.
- Machine Learning would only be used to help the robot learn from its environment or past experiences.

Module 1:

How Does AI Work?



- **Study Large Data:** AI analyzes large amounts of data to find patterns.
- **Make Predictions:** AI makes predictions based on the data it has studied.
- **Autonomous Decision Making:** AI can make decisions on its own, learning from past data and experiences.
- **Adapt and Improve:** AI systems continuously adapt and learn from new data to improve over time.
- **React to Problems:** AI systems perceive problems and respond accordingly, using the patterns they have learned.
- **Data-Driven:** AI relies on large amounts of data to make decisions, with cheap data storage and fast processors making it easier to use.
- **Make Accurate Predictions:** AI can make accurate predictions based on past data and experience.
- **Growing and Evolving:** AI is being applied in many different areas and its scope is continuously expanding.

Module 1:

Defining Intelligence



Linguistic Intelligence

The ability to speak and understand language effectively, like narrators and orators demonstrate.



Spatial Intelligence

The ability to visualize and manipulate images mentally, like map readers and astronauts do.



Interpersonal

Understanding others' feelings and intentions, exemplified by mass communicators and interviewers.



Musical Intelligence

The ability to create and understand music, recognizing rhythm and pitch like musicians and singers.



Bodily-Kinesthetic

The ability to use your body to solve problems, seen in dancers and athletes.



Logical-Mathematical

The ability to understand abstract concepts and use logic, exemplified by mathematicians and scientists.



Intrapersonal

Understanding your own feelings and motivations, like spiritual leaders and philosophers.

Artificial Intelligence (AI): A system or machine is considered artificially intelligent if it can exhibit one or more of these types of intelligence.

Module 1:

Components of Intelligence

- Reasoning
- Learning
- Problem solving
- Perception
- Linguistic intelligence

Module 1:

Components of Intelligence - 1. Reasoning

Reasoning is the process used to make decisions and predictions. It involves analyzing information and drawing conclusions based on evidence. There are two main types of reasoning:

- **1. Inductive Reasoning:**

- What it is: Making generalizations based on specific observations or examples.
- How it works: Starts with specific observations and moves to a general conclusion.
- Example: If you see 10 white swans, you might conclude that all swans are white.

- **2. Deductive Reasoning:**

- What it is: Drawing a specific conclusion based on general principles or facts.
- How it works: Starts with a general statement or premise, and moves to a specific conclusion.
- Example: Tom is a cat, All cats are mammals, Hence Tom is a mammal.

Module 1:

Components of Intelligence - 1. Reasoning

I observe that every dog I meet on my street is friendly, hence I conclude that all dogs are friendly. (Inductive Reasoning)

Five friends who bought cars from Tesla are satisfied, so the company likely makes reliable car. (Inductive Reasoning)

If a person is born between 1997 and 2012 then he/she/they are GEN -Z. ... was born in 2007; therefore, ... is Gen Z. (Deductive Reasoning)

The freezing point of water is 0°C . It's hotter than 0°C , so water will not freeze. (Deductive Reasoning)

Carrots are orange; oranges are orange; therefore, oranges are carrots.

Module 1:

Components of Intelligence - 2. Learning

Learning is the process of gaining knowledge or skills by studying, practicing, or experiencing something. It helps humans, animals, and even AI systems to improve their understanding of different subjects.

There are different types of learning:

1. Auditory Learning:
2. Episodic Learning:
3. Motor Learning:
4. Observational Learning:
5. Perceptual Learning:
6. Relational Learning:
7. Spatial Learning:
8. Stimulus-Response Learning:

Module 1:

Components of Intelligence - 2. Learning



 Auditory Learning Learning by hearing and listening. <i>Example:</i> Listening to recorded lectures and concepts.	 Episodic Learning Learning by remembering personal experience, memories. <i>Example:</i> Recalling recipe steps in order.
 Motor Learning Learning through physical movement. <i>Example:</i> Learning proper object handling techniques.	 Observational Learning Learning by watching others. <i>Example:</i> Children copying their parents' actions.
Perceptual Learning Recognizing familiar things. <i>Example:</i> Wine experts learning to distinguish subtle flavor differences.	Relational Learning Recognizing relationships and connections. <i>Example:</i> Adjusting spice amounts based on previous cooking experience.
Spatial Learning Learning through visual stimuli. <i>Example:</i> Creating mental route maps before driving.	Stimulus-Response Learning Learning through reactions. <i>Example:</i> Shouting when touching a hot pan due to pain response.

Learning Type	Auditory	Episodic	Motor	Observational	Perceptual	Relational	Spatial	Stimulus-response
Human Example	Toddler learning to recognize spoken instructions from parents	Remembering to avoid touching a hot surface after getting burned once	Person learning to drive a car by practicing steering and brake control	Child learns to swim by watching a sibling	Birdwatcher distinguishing bird species by subtle visual differences	Student understanding that "adding fuel increases a cars speed	Learning to navigate a new city after walking around	Dog learns to sit when hearing the command after rewards
AI Example	Speech recognition AI learning to understand different dialects	Chatbot adjusting its replies based on past conversation errors	Robotics arms improving manipulation skills through trial and error in manufacturing	AI learning to play video games through observing human player strategies	Image recognition AI distinguishing objects in complex scenes	Knowledge graph AI linking entities like "Einstein" with "physicist"	Autonomous vehicle mapping optimal routes using sensor data	Reinforcement learning AI trained to perform specific actions based on rewards

Module 1:

Components of Intelligence - 3. Problem Solving

Problem solving is the process of finding a solution to a challenge or issue. It involves:

Identifying the problem: Understanding the situation and recognizing the obstacles (either known or unknown).

Making decisions: Choosing the best approach or method to overcome the obstacles and reach the goal.

- Creativity
- Intuition
- Experience



Module 1:

Components of Intelligence - 4. Perception

Perception is the process of:

- Acquiring information through the senses (like sight, hearing, etc.).
- Interpreting that information to understand what's happening around us.
- Selecting important details and organizing them to form a clear picture.
- Humans use sensory organs (like eyes, ears) to perceive the world.
- AI systems use sensors (like cameras, microphones) to gather data and understand their environment.

Components of Intelligence - 5. Linguistic Intelligence

It is used in interpersonal communication and defines one's ability to use, comprehend, speak and write the verbal and written language

Module 1:

Differences Between Human and Machine Intelligence

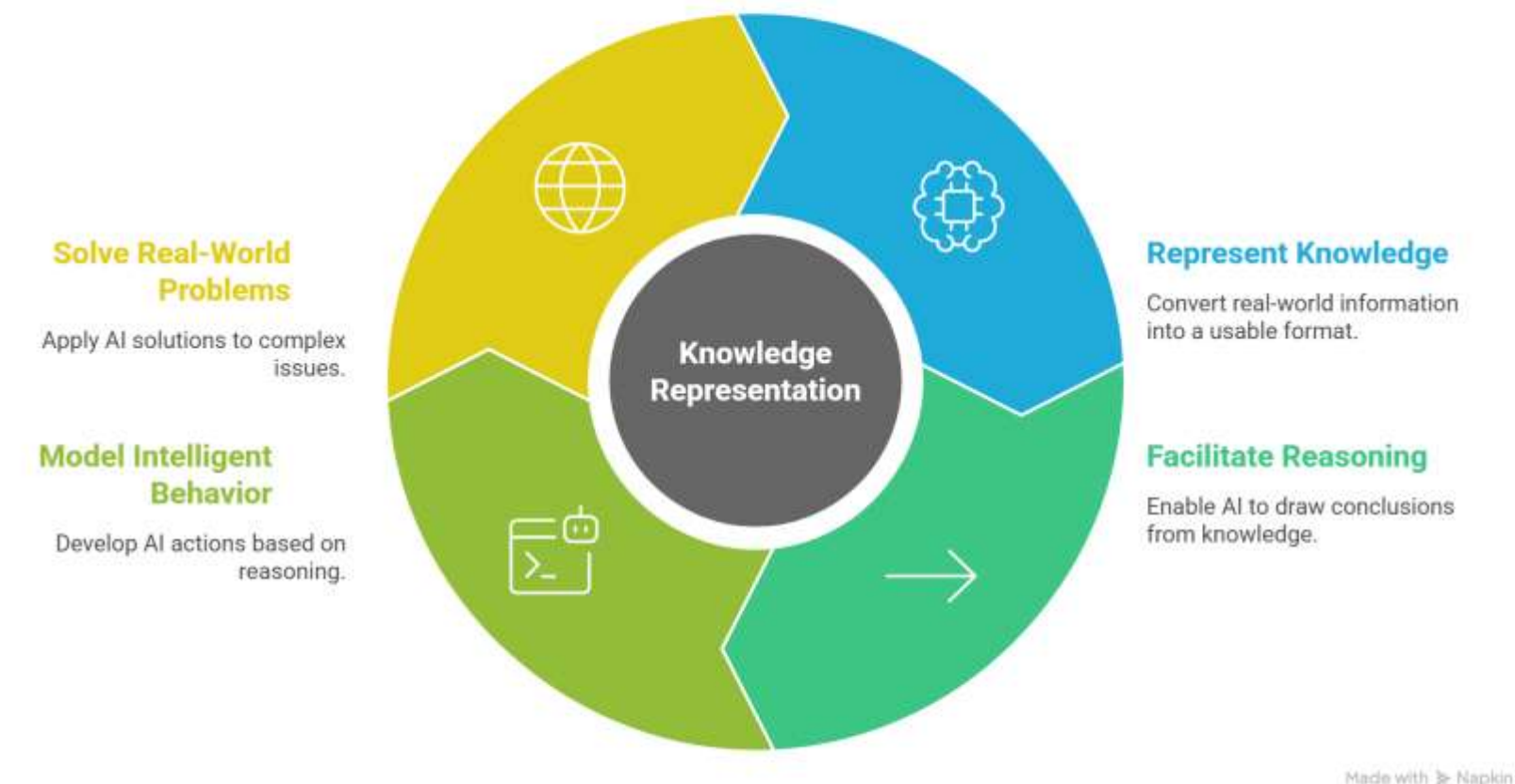
Aspect	Human Intelligence	Machine Intelligence
Perception	Perceives through patterns	Perceives by analyzing data with rules
Memory and Recall	Recalls information by patterns	Uses search algorithms to find information
Handling Missing Information	Can deduce missing or distorted information accurately	Struggles with incomplete data, less accurate

Module 1:

Knowledge Representation

Introduction

- Artificial intelligence as technology has always fascinated human beings. Multiple science fiction novels and movies have demonstrated the use of AI-powered systems (like robots) that can think, act, understand complex information and make smart decisions based on it.
- Although this all looks like a fantastic development, we agree that it is not easy to make machines that behave exactly like humans. This is because humans have conscience and this conscience develops gradually in us with our knowledge, experiences and memories.
- To build AI systems that have conscience, we need to inculcate knowledge in them. This chapter is, therefore, dedicated towards representing knowledge in machines.



Module 1:

Knowledge Representation

Knowledge and Intelligence

- Knowledge of real world plays a crucial role in intelligence and creating AI agents that demonstrate intelligent behaviour. Such an agent acts by sensing its environment and using knowledge to act intelligently.
- However, for this, the knowledge or experience about the input is mandatory. The relationship between knowledge and intelligence can be clearly understood from the figure. You can see that if we remove the knowledge component, then the decision maker will not be able to sense the environment accurately and take appropriate decisions



Relationship between Knowledge and Intelligence

Humans use knowledge to understand, reason, and act.

But machines? For them, this comes under the domain of **Knowledge Representation and Reasoning (KRR)**.

KRR is about:

- How AI machines *think*,
- How their thinking leads to intelligent behaviour,
- And how they can solve problems like diagnosing diseases or chatting with in natural language.

Module 1:

Knowledge Representation



What Knowledge Needs to be Represented?

Humans have intuition, intentions, prejudices, beliefs, judgments, common sense, etc. apart from knowledge about certain facts. We need to incorporate all this information in a machine-understandable format and make the AI system truly intelligent. For this, we need to represent the following knowledge in AI systems:

Object: Information and facts about all objects (relevant in context). For example, in a self-driving car, vehicles and roads are objects.

Events: Information and facts about actions which occur in the real world. For example, in a self-driving car, an event can be applying breaks when an object comes in front of it.

Performance: The manner in which actions are performed. It describes the behaviour related to how to do things.

Meta-knowledge: It is the knowledge about knowledge (what we know).

Facts: Facts are the truths about the real world that needs to be represented for an intelligent agent.

Knowledge base: It is the most important component of a knowledge-based agent that stores a group of sentences (technical sentences, not simple English language ones)

Type	Medical AI	Self-driving Car	Netflix
Objects	Symptoms, organs, diseases, patients	Roads, vehicles, pedestrians, traffic lights	Movies, users, genres, actors
Events	Patient develops fever" → Check for infections	Red light appears" → Apply brakes	User pauses romantic movie" → Maybe suggest comedy next time?
Performance	How to prioritize symptoms by severity	How to merge safely into traffic	How to balance popular vs. personalized recommendations
Meta-knowledge	I'm more confident about common diseases than rare ones	My vision is impaired in heavy rain	This user's ratings are more reliable than their viewing time
Facts	Aspirin reduces fever	Speed limit on Highway 101 is 65 mph	This movie was released in 2019

Knowledge Base: SQL, AWS, Apache Cassandra, Influx DB (Tesla)

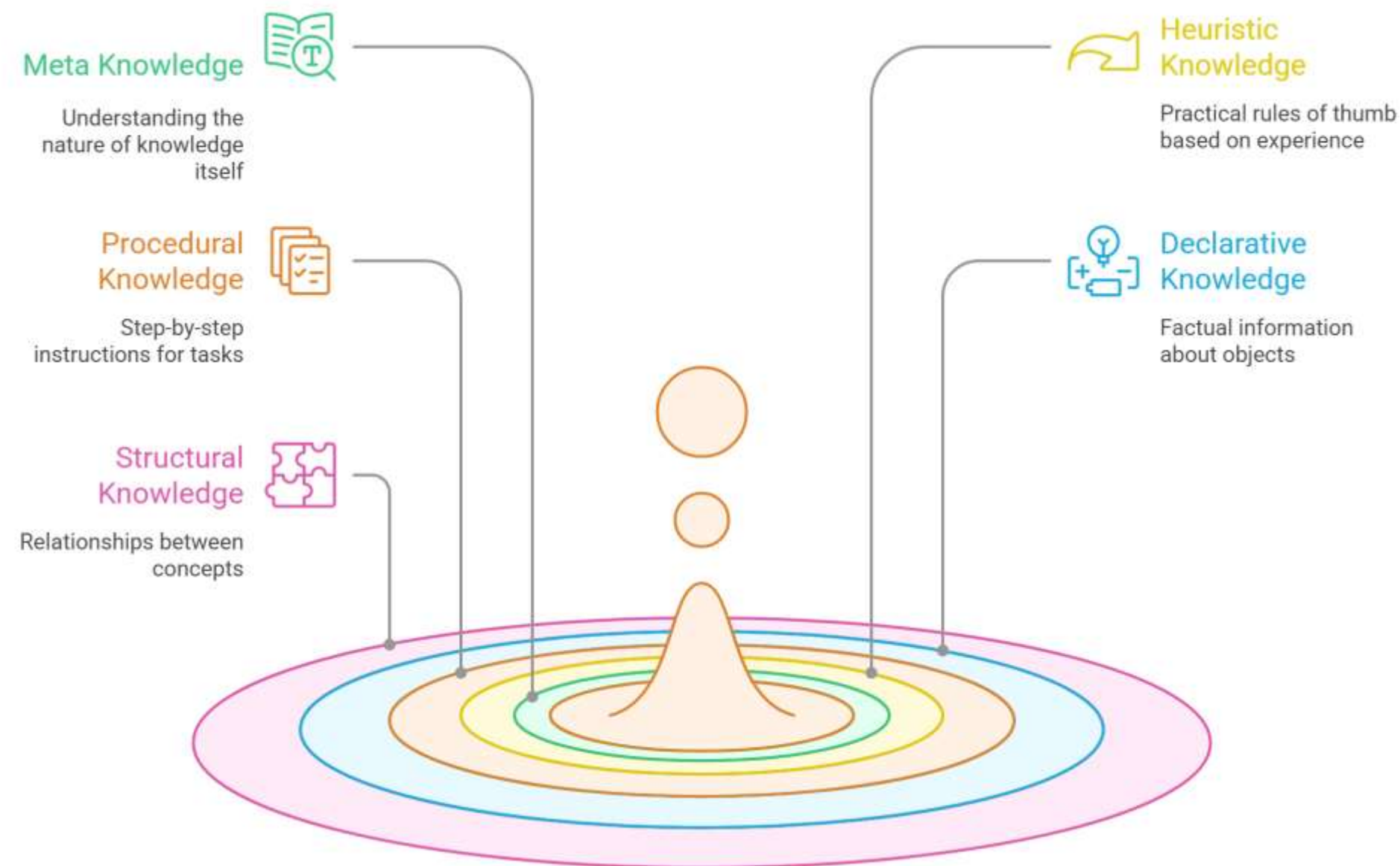
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What is Knowledge?

We know that knowledge is the basic element of our brain as it helps us to know and understand the things logically and is gained by experiences of facts, data and situations. A knowledgeable person performs all the actions in a better way. Everyone has five types of knowledge

- **Meta knowledge** is the knowledge about knowledge.
- **Heuristic knowledge** is the knowledge about a specific topic. For example, it can be knowledge of some experts in a field or subject. Heuristic knowledge is treated as the rule of thumb as it is based on previous experiences and awareness of approaches, which are good to work but not guaranteed.
- **Procedural knowledge**, also known as imperative knowledge, gives information about how to achieve or do something. This knowledge includes rules, strategies, procedures, agendas, etc. that can be directly applied to perform any task.
- **Declarative knowledge** is the information that we have about an object. This knowledge helps us to describe a particular concept, fact, object and its attributes. Declarative knowledge is simpler than procedural language and is also called descriptive knowledge as it is usually represented using declarative sentences.

- **Structural knowledge** is the basic knowledge to solve complex problems. It describes relationships between various concepts or objects such as kind of, part of, and grouping of something.



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Declarative knowledge



Definition

Facts, concepts, and "what is" information.



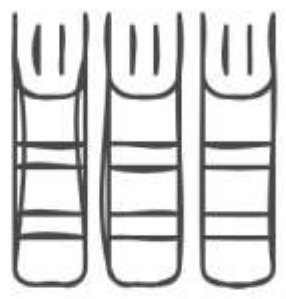
Characteristics

Explicit, can be stated directly, theoretical understanding.



Examples

Paris is capital of France, photosynthesis definition, periodic table.



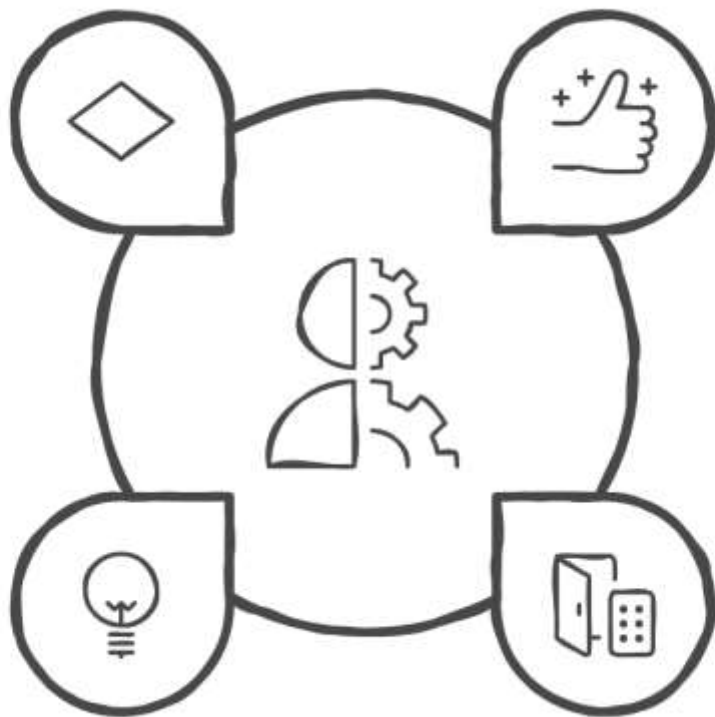
Visual Concept

Books, encyclopedias, fact sheets.

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Visual concept

Compass, lightbulb moments, decision trees



Examples

Keep it simple," diagnostic guidelines, strategic approaches

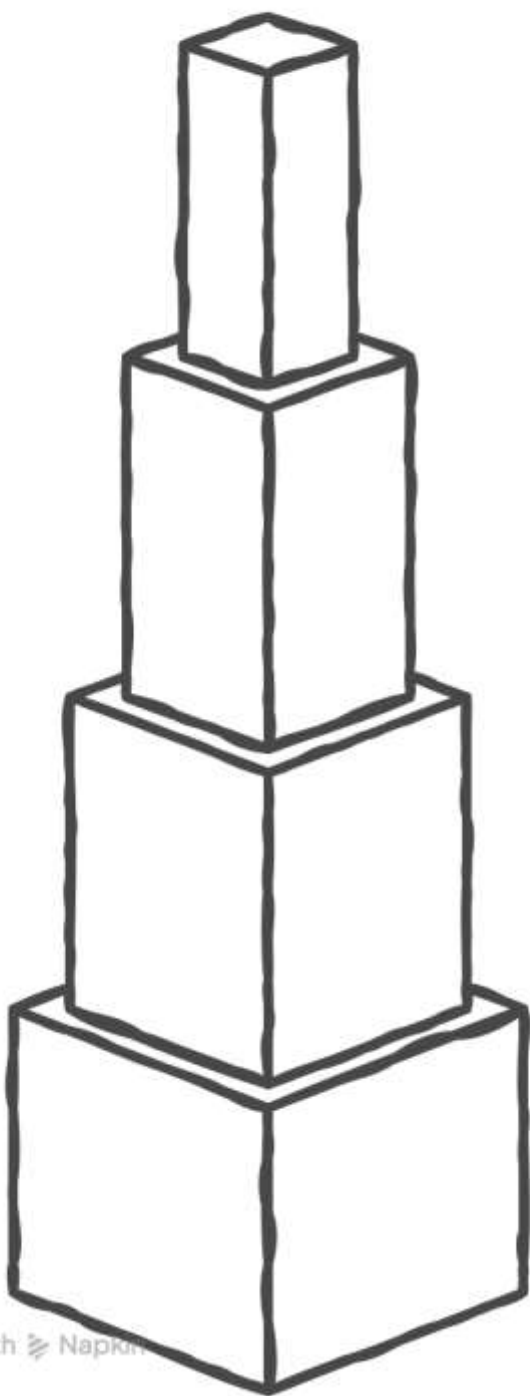
What it is

Rules of thumb and practical problem-solving shortcuts

Key characteristics

Experience-based, guides decisions, handles uncertainty

Heuristic knowledge



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Structural knowledge

What it is

Understanding relationships and organization between concepts

Key characteristics

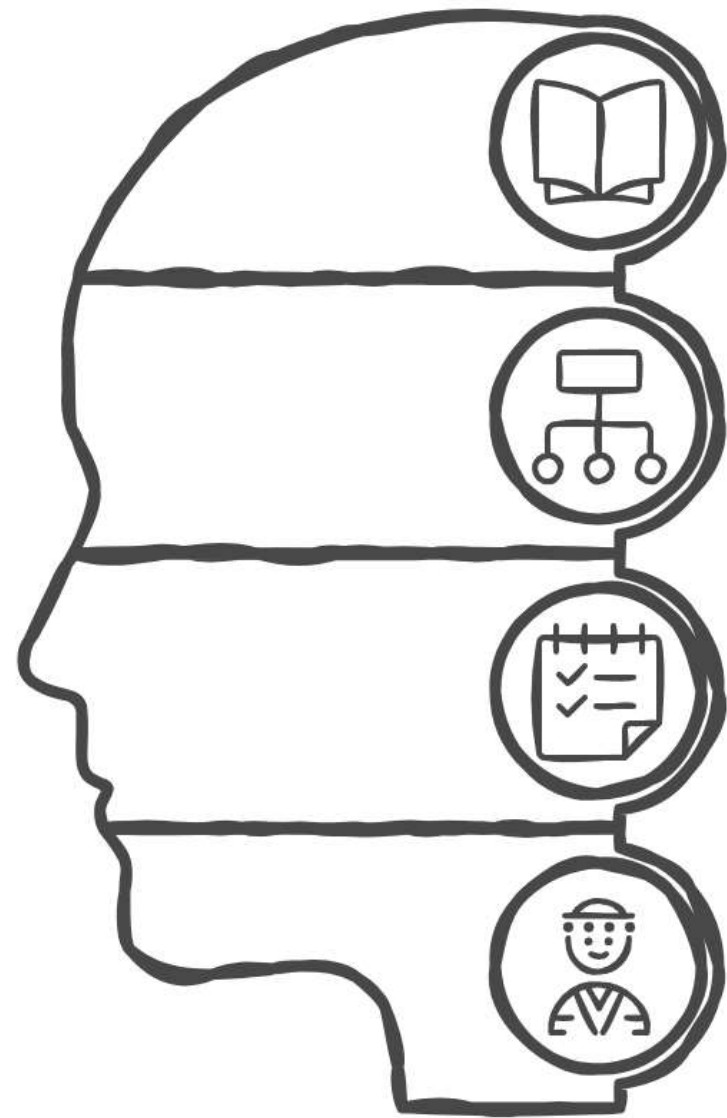
Shows connections, patterns, hierarchies, frameworks

Examples

How body systems connect, historical cause-and-effect, theory components

Visual concept

Mind maps, organizational charts, network diagrams



Definition

Knowledge about your own knowledge and thinking

Key Aspects

Self-awareness, monitors learning, regulates cognition

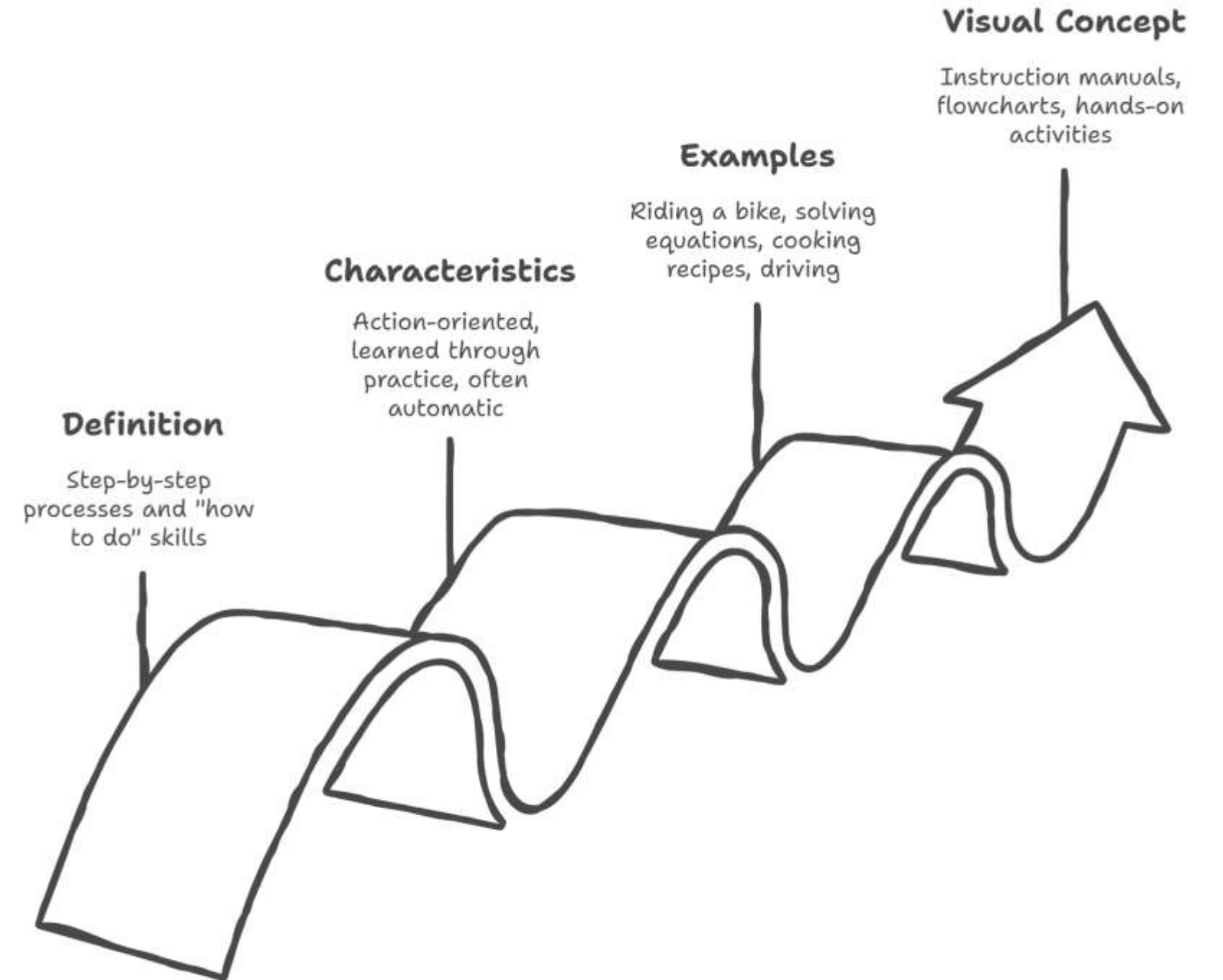
Practical Use

Knowing your limits, choosing study strategies, recognizing biases

Visual Representation

Mirror reflecting brain, control dashboard, self-reflection

Meta knowledge



Procedural knowledge

Characteristic	Definition	Examples	Application
Declarative	What the chef knows about biryani	Rice varieties, spice properties, ingredient facts	Aged basmati absorbs flavors better
Procedural	What the chef does to make biryani	Step-by-step cooking, knife techniques, spice tempering	Marinate meat, cook rice, layer
Structural	What the chef understands about biryani	How flavors build, relationship between cooking time and texture	Mint loses freshness if added wrong
Heuristic	What the chef intuitively knows about biryani	Rules of thumb from experience, shortcuts	Sound indicates moisture is right
Meta	What the chef knows about their biryani skills	Self-awareness of abilities, when to trust instinct vs recipe	Gas stove runs high, use tawa

Module 1:

Knowledge Representation



What is Logic?

Logic is the main component of any knowledge as it facilitates a person to draw conclusion by filtering required information from large volumes of sentences.

In AI, knowledge is represented using logic. Every logic has three main elements.

Syntax

Every language has its own syntax to specify the sequence of constructs that makes a complete sentence in that language. It is therefore not wrong to say that syntax is the representation of a language.

Semantics

Semantics is used to check if a syntactically correct sentence is logically correct and meaningful or not. Therefore, semantics defines the sense of the sentence which relates to the real world.

For example, consider the statement:

Ram is riding a bike.

This sentence is syntactically as well as semantically correct. However, the sentence, *Bike is riding Ram.* is syntactically correct but semantically incorrect

Logical Inference

Inference means to deduce conclusions in the context of some fact or problem. Correspondingly, logical inference uses inference algorithms to think all the possible reasons that can give a proper result

Module 1:

Knowledge Representation

Cycle of Knowledge Representation in AI

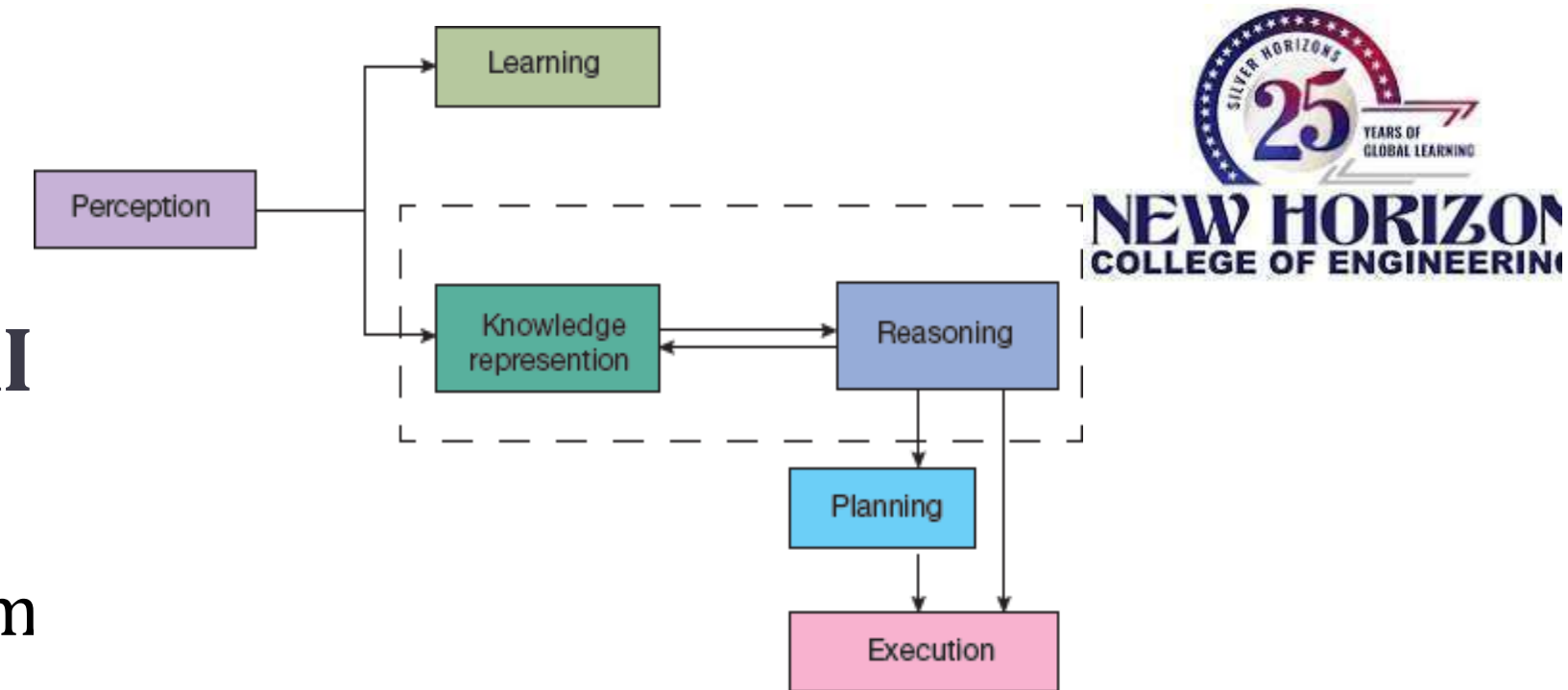


FIGURE 4.3 Components of an AI system

An AI system has several components ([Fig. 4.3](#)) that helps it to exhibit an intelligent behaviour. These components include the following:

Perception

The perception component retrieves data from the environment, discovers the source(s) of noise, checks if the AI was damaged by anything and defines response to be given when any sense has been detected. Data from the environment is gathered using different types of sensors in the form of video, audio, text, time, temperature or any other sensor-based input.

Learning

The learning component learns from the data that is captured by the perception component. The aim here is to develop a system that can be taught instead of being programmed. With learning, the system focuses on self-improvement through knowledge acquisition, inference, acquisition of heuristics, faster searches, etc.

Knowledge Representation and Reasoning

This is the main component which shows human-like intelligence in the machines. Knowledge representation techniques (refer [Fig. 4.4](#)) are designed to understand using a top-down approach focussing in detail, what an agent needs to know to behave intelligently. The KRR component also defines how automated reasoning procedures can make this knowledge available as needed



FIGURE 4.4 Forms of Knowledge Representa

Planning and Execution

This component analyses knowledge representation and reasoning. The planning component selects an initial state, enumerates preconditions and effects, and a sequence of actions to achieve the goal state. Once planning is done, actions are executed to get the desired results

Knowledge Representation Requirements

A good knowledge representation system must have the following properties.

1. Representational accuracy, to ensure that the system represents all kinds of required knowledge.
2. Inferential adequacy to manipulate the representational structures to produce new knowledge corresponding to the existing structure.
3. Inferential efficiency to direct the inferential knowledge mechanism to generate appropriate results.
4. Acquisitional efficiency that helps the knowledge representation system to easily acquire new knowledge using automatic methods.

Cycle of Knowledge Representation in AI

[SELF DRIVING CAR]

Perception

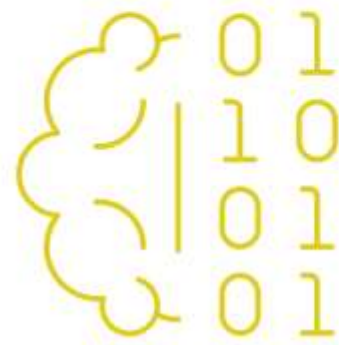


Car detects a red traffic light 50 meters ahead, a pedestrian crossing on the right, and a vehicle braking in the adjacent lane.

Cycle of Knowledge Representation in AI

[SELF DRIVING CAR]

Learning



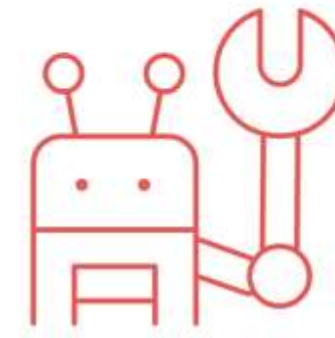
What it does

Car's AI learns from all the perception data to improve driving performance.



Learning focus

Pattern recognition (stop signs vs speed limit signs), behavior prediction (pedestrian likely to cross), route optimization.



Self-improvement

Gets better at recognizing objects in different weather, learns local traffic patterns, adapts to new road conditions.



Knowledge acquisition

Learns that school zones mean slower speeds, construction zones require lane changes.

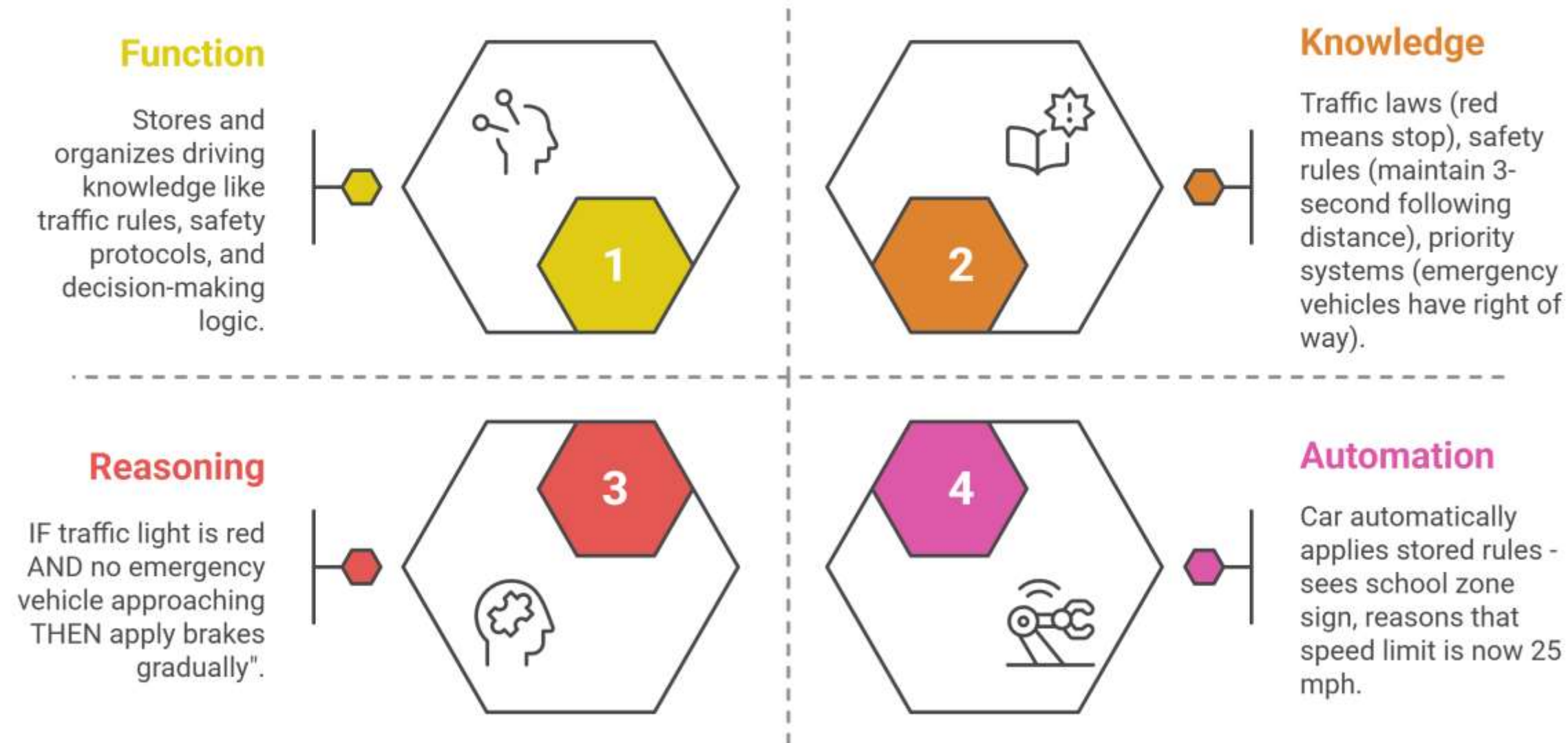
Made with  Napkin

After 1000 hours of driving, the AI learns that pedestrians near bus stops are more likely to suddenly enter the road.

Cycle of Knowledge Representation in AI

[SELF DRIVING CAR]

Knowledge Representation and Reasoning (KRR)

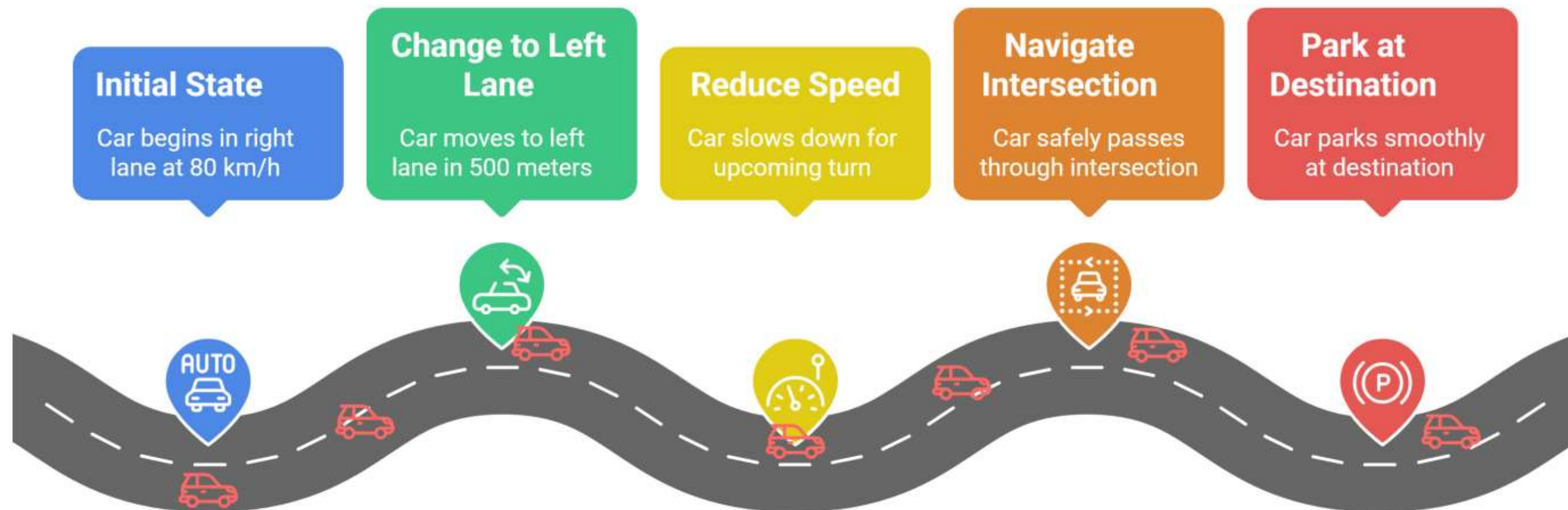


Made with  Napkin

The car recognizes ambulance with sirens, accesses emergency vehicle protocol, reasons it must safely move to shoulder.

Cycle of Knowledge Representation in AI

[SELF DRIVING CAR]



Module 1: Introduction to Artificial Intelligence



1. Introduction to AI

1. Definition of Artificial Intelligence
2. How Does AI Work?
3. Advantages and Disadvantages of Artificial Intelligence
4. History of Artificial Intelligence
5. Types of Artificial Intelligence:
 - a. Weak AI vs. Strong AI
 - b. Reactive Machines
 - c. Limited Memory
 - d. Theory of Mind
 - e. Self-Awareness
6. Is Artificial Intelligence the Same as Augmented Intelligence and Cognitive Computing?

2. Machine Intelligence

1. Defining Intelligence
2. Components of Intelligence
3. Differences Between Human and Machine Intelligence
4. Knowledge Representation and its types

Module 1: Introduction to Artificial Intelligence



MODEL QUESTIONS

1. Define AI and give advantages and disadvantages of this technology.
2. How do AI systems work?
3. Differentiate between Weak AI and strong AI
4. Differentiate between Reactive machines and limited memory AI systems
5. Differentiate between Augmented and artificial intelligence
6. How will you categorize AI systems based on their functionality?
7. Briefly discuss the History of AI
8. Define the types of intelligence
9. Explain the components of intelligence
10. Differentiate between human and Machine Intelligence
11. Discuss the relationship between knowledge and intelligence
12. Discuss the knowledge to be represented in AI systems
13. Explain the types of Knowledge
14. Explain the cycle of knowledge representation in AI and the requirements

Thank You

Module 1 - Completed